



Ocimum basilicum attenuates ethidium bromide-induced cognitive deficits and pre-frontal cortical neuroinflammation, astrogliosis and mitochondrial dysfunction in rats

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Abstract

Multiple sclerosis (MS) is a chronic neurodegenerative disorder with clinical symptoms of neuroinflammation and demyelination in the central nervous system. Recently, herbal medicines are clinically effective against MS as the current disease-modifying drugs have limited effectiveness. Hence, the present study evaluated the therapeutic potential of *Ocimum basilicum* essential oil (OB) in ethidium bromide (EB)-induced cognitive deficits in the male rats. Further, the effect of OB (50, 100 and 200 $\mu\text{L/kg}$) was evaluated on EB-induced neuroinflammation, astrogliosis and mitochondrial dysfunction in the pre-frontal cortex (PFC) of the animals. The EB was injected through bilateral intracerebroventricular route into hippocampus to induce MS-like manifestations in the rats. OB (100 and 200 $\mu\text{L/kg}$) and Ursolic acid (UA) significantly reduced the EB-induced cognitive deficits in Morris water maze and Y-maze test paradigms. OB (100 and 200 $\mu\text{L/kg}$) and UA significantly attenuated the EB-induced neuroinflammation in terms of increase in the levels of pro-inflammatory cytokines (TNF- α and IL-6) in the rat PFC. Further, OB (100 and 200 $\mu\text{L/kg}$) and UA significantly attenuated the EB-induced astrogliosis in terms of increase in the levels of GFAP (Glial fibrillary acidic protein) and Iba-1 (Ionized calcium binding adaptor molecule-1) in the rat PFC. In addition, OB (100 and 200 $\mu\text{L/kg}$) and UA significantly attenuated the EB-induced decrease in the mitochondrial function, integrity, respiratory control rate and ADP/O in the PFC of the rodents. Moreover, OB (100 and 200 $\mu\text{L/kg}$) and UA significantly reduced the EB-induced mitochondria-dependent apoptosis in the PFC of the rat. Hence, it can be presumed that OB could be a potential alternative drug candidate in the pharmacotherapy of MS.

Keywords *Ocimum basilicum* · Astrogliosis · Hippocampus · Mitochondria · Multiple sclerosis · Neuroinflammation

Introduction

Multiple sclerosis (MS) is a chronic debilitating neurodegenerative disease which includes cardinal features such as relapses, remissions and progression of disability of central nervous system activity over time (Compston and Coles 2002). The cognitive impairment especially in declarative memory is observed in more than 70% of patients with MS (Julian 2011; González Torre et al. 2017). The severity and type of cognitive impairment in

MS varies considerably among individuals and can be observed in early and in later stages of the disease (Feinstein 2015; D'Amico et al. 2016). The prominent hypothesis of pathophysiology of MS is immune cell-induced demyelination in the neurons of the central nervous system. It has been well documented that cholinergic dysfunction occurs in the demyelinating lesions in MS and the drugs that improve the cholinergic activity also enhance the cognitive functions in MS (Porcel and Montalban 2006). However, these drugs are restricted due to their severe adverse effects. Hence, a novel and alternate medicine should be developed in the management of cognitive deficits in MS. Preclinical studies suggest therapeutic effectiveness of several herbal medicines in the management of cognitive impairment in MS (Ghaffari et al. 2015; Carvalho et al. 2015) and thus gains critical attention in the management of MS.

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Ocimum basilicum Linn (family Lamiaceae; OB), commonly known as Sweet Basil, has been used for culinary purposes for hundreds of years throughout Italy and all over the world in salads, tomato based dishes and in pasta sauces such as the Italian pesto. It is a component of several food supplements, which are easily accessible as products able to maintain and promote health. They are sold over the counter and highly accepted by the consumers. OB is reported to have therapeutic activity against neurological disorders such as depression, anxiety and sedation. It is also suggested that the therapeutic effects of OB is mostly due to its anti-inflammatory and antioxidant property (Sestili et al. 2018). Several studies have been carried out to investigate the therapeutic potential of OB and its active constituent ursolic acid (UA) against different animal models of cognitive deficits (Liang et al. 2016; Singh et al. 2018; Ayuob et al. 2018). However, the therapeutic effectiveness of OB against memory impairment in animal model of MS is yet to be proven.

Local injection of ethidium bromide (EB) to the brain region of the laboratory animals is a simple experimental model of MS (Blakemore 1982; Bondan et al. 2006). Neuroinflammation is considered as one of the predisposing factor in the pathophysiology of MS (Goodwin 2016). It is well suggested that EB can cause demyelination through neuroinflammation in the hippocampus of the rodents (Goudarzvand et al. 2016). Recent imaging studies demonstrate the degenerations of the neurons of the other brain regions including cortex in MS patients (Cuzzo 2005; Sicotte et al. 2008; Roosendaal et al. 2009). It has been well suggested that the activation and proliferation of microglia and astrocytes can cause neuroinflammation within the demyelinating lesions in MS (Zielasek et al. 1992; Kurkowska-Jastrzebska et al. 1999; Mayo et al. 2012). In this context, it has been suggested that astrogliosis can be a potential target to reduce the reactive oxygen species formation and neuroinflammation in MS (Bondan et al. 2006; Jamby and Sofroniew 2010). It is well documented that the extent of mitochondrial dysfunction is usually observed in astrogliosis and neuroinflammation (Lott et al. 2007). Moreover, mitochondrial dysfunction is also reported along with neuroinflammation and astrogliosis in the brain of the animals that are challenged to experimental autoimmune encephalomyelitis (Niinobu et al. 2000; Lee et al. 2018). Therefore, mitochondria could be a potential target along with neuroinflammation and astrogliosis to ameliorate the cognitive deficits in the management of MS.

Hence, the present study evaluated the therapeutic potential of OB in EB-induced cognitive deficits in the animals. Further, the effect of OB was evaluated on EB-induced neuroinflammation, astrogliosis and mitochondrial dysfunction in the pre-frontal cortex (PFC) of the animals.

Materials and methods

Animals

Male adult wistar albino rats (180 ± 20 g) were procured from the central animal house, Institute of Pharmaceutical Research, GLA University, Mathura. Forty two animals in experimental study were divided into seven groups of six each. Rats were housed in poly-ethylene cage at ambient temperature 25 ± 1 °C and of relative humidity 45–55%, with a 12:12 h light/dark cycle. Animals were freely allowed to feed their standard pellet (Lipton India, Ltd., Mumbai) and water ad libitum during the experiment. The experimental procedures were performed according to the principles of the Institutional Animal Ethics Committee (1260/PO/ERe/S/09/CPCSEA/IAEC/2019/P.Col/R.20) under strict compliance of the Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA) guidelines for experimental studies. Further, principles of laboratory animal care (National Research Council (US) Committee for the update of the Guide for the Care and Use of Laboratory Animals 2011) were followed during the experiments.

Chemicals and reagents

The chemicals such as ethidium bromide (EB), TMRM, *Ocimum basilicum* essential oil (Basil Oil) and Ursolic acid (UA) were procured from Sigma (St. Louis, MO, USA). All other analytical grade reagents and chemicals were purchased from Merck Pvt. Ltd., New Delhi.

Stereotaxic injection of EB

The scalp of anesthetized rat (sodium pentobarbital; 40 mg/kg; i.p.) was incised and retracted, and the head was positioned to place bregma and lambda in the same horizontal plane in a stereotaxic frame (Stoelting, USA). One small hole of 1.5 mm depth was drilled on right side of dentate gyrus of hippocampus and stainless-steel guide cannula was implanted (posterior – 2.8 mm, medial/lateral ± 1.8 mm relative to bregma, dorsal/ventral – 2.5 mm below dura (Paxinos and Watson 1998). EB, dissolved in artificial cerebrospinal fluid (aCSF), was injected through Hamilton syringe via polyethylene tube ($10 \mu\text{g}/\mu\text{L}$, $3 \mu\text{L}/\text{animal}$, $1.0 \mu\text{L}/\text{min}$) into hippocampus (Goudarzvand et al. 2016). In Sham animals three μL of aCSF (147 mM NaCl, 2.9 mM KCl, 1.6 mM MgCl_2 , 1.7 mM CaCl_2 and 2.2 mM dextrose; pH – 7.4) was infused into the hippocampus. All microinjections were done with Quintessential Stereotaxic Injector (Stoelting, USA).

Experimental protocol

The whole experimental protocol was designed for 14 days (Fig. 1). Animals were acclimatized for 7 days and then randomly divided into 7 groups each containing 6 animals namely Control, Sham, EB, EB + OB-50, EB + OB-100, EB + OB-200 and EB + UA. EB (10 µg/µL) was administered on Day-1 (D-1) of experimental schedule to all animals except control and sham group rats. OB (50, 100 and 200 µL/kg, p.o.; Rashidian et al. 2016) and UA (5 mg/kg, p.o.; Wang et al. 2016) were administered to EB + OB-50, EB + OB-100, EB + OB-200 and EB + UA group animals from D-1 to D-14 of experimental schedule respectively. The fresh emulsion of OB extract was daily prepared in 0.5% (v/v) of Tween-80 in normal saline and administered per oral to the animals. Drugs were administered for 14 consecutive days to evaluate the neuroprotective potential in EB-induced cognitive deficits in rats. Animals of Control group were treated with Tween-80. As the protocol was scheduled for 14 days, Morris water maze (MWM) test was performed for five consecutive days (from D-10 to D-14) to evaluate the learning and memory formation of the animals. Subsequently, Y-maze test was performed on D-14 after 1 h to MWM test paradigm to evaluate the spatial memory formation in the animals. All behavioral parameters were recorded and quantified with ANY-maze™ (Version-4.96, USA) video-tracking system. After behavioral observation animals were sacrificed by decapitation method. The brain was microdissected (Paxinos and Watson 1998) into PFC and stored immediately at -80 °C for biochemical estimation. All the biochemical experiments were repeated twice for reproducibility.

Assessment of effect of OB (50, 100 and 200 µL/kg) on EB-induced cognitive deficits in different behavioral models

Morris water maze test paradigm

The standard protocol of MWM test procedure was performed to assess learning and memory function of the animals (Morris 1984; Pandey and Garbhadu 2017). Briefly, task includes a circular tank painted whole through black and filled with water at the level of 30 cm. Diameter and height of tank was 150 cm and 60 cm respectively. Temperature of tank water was maintained at 27 ± 1 °C. Pool was divided into four equal quadrants. A hidden platform of 4.5 cm in diameter was formed in the middle portion of target quadrant and settled 1.5 cm beneath the upper layer of water. Action of animals was recorded through a video tracking system placed over the water tank. Randomly rats were positioned in the water tank in any one of the quadrants for 120 s and made them free to swim and find out the hidden platform. If animals were unable to find out the platform then they were trained to grasp the platform. The escape latency to find out the hidden platform of all groups of animals was observed and the time taken to find out the platform was recorded by the software. Each rat was placed four times between the gaps of 5 min in each trial to observe the escape latency of them. On fifth day, the hidden platform was removed from the tank. Time spent in each quadrant and time spent in the target quadrants to find out hidden platform was noted down. Percentage of total distance travelled in target quadrant and swimming speed of all groups of animals was also noted down. Time

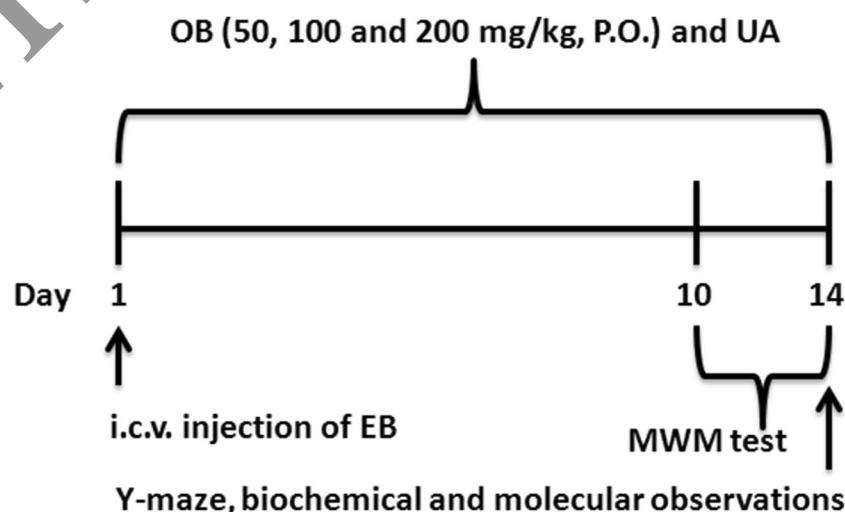


Fig. 1 Diagrammatic representation of detailed experimental schedule

spent in target quadrant was considered as mark of memory rejuvenation.

Y-maze test paradigm

The working memory was determined by observing spontaneous alteration behavior (SAB) through Y-maze test paradigm (Mouri et al. 2007). Test was conducted on last day (14th) of experimental protocol. Y-maze apparatus includes 3 arms (40 cm long and 3 cm wide, with walls 12 cm high) painted whole through black color (named as A, B, C) disposed at 120° apart. Rat was positioned at midpoint of the apparatus and made them free to move. Total number of arm entries and sequences of consecutive entries (ABC, CAB, or BCA, but not BAB) were recorded for 8 min. Arms were wiped throughout the trails to remove resting odor. The percentage alteration behavior was calculated using equation, % spontaneous alteration behavior = [(total number of alternations) / (total number of arm entries – 2)] × 100.

Estimation of IL-6 and TNF- α

The level of IL-6 and TNF- α were estimated spectrophotometrically at 450 nm using an enzyme-linked immunosorbent assay kit (Abcam Pvt. Ltd., India) according to the manufacturer's instructions.

Assessment of effect of OB (50, 100 and 200 mg/kg) on EB-induced brain mitochondrial dysfunction

Isolation of mitochondria from rat PFC

The standard protocol of Pedersen et al. (1978) was used to isolate the mitochondria from each tissue. The mitochondrial protein content was estimated in each tissue fraction using standard method of Lowry et al. (1951).

Estimation of mitochondrial function

The (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) reduction assay was considered to assess mitochondrial function in each tissue fraction in terms of estimating the level of formazan formed using spectrophotometric method at 595 nm wavelength (Kamboj et al. 2008). The results were expressed as mg formazan formed/min/mg protein.

Evaluation of MMP in PFC

The MMP was estimated in terms of the extent of rhodamine dye taken up by mitochondria was measured in spectrofluorometer (Hitachi, F-2500) at an excitation λ of 535 ± 10 nm

and emission λ of 580 ± 10 nm (Huang 2002). The results were expressed as fluorescence intensity/mg protein.

Estimation of mitochondrial bioenergetics

Clark oxygen electrode (Hansatech Instrument Pvt. Ltd., USA) was used to monitor the extent of mitochondrial respiration using principle of polarography. 1 mg/ml of isolated mitochondria preparation was incubated in a respiration medium containing 125 mM sucrose, 65 mM KCl, 2.5 mM $MgCl_2$, 5 mM KH_2PO_4 , 5 mM Hepes, pH 7.2 at 30 °C. An initial rate of oxygen consumption (state 2 or V_2) was recorded after addition of pyruvate plus malate (10 mM/5 mM). Subsequently, the state 3 rate (V_3) was recorded after addition of 250 nmol of ADP. Thereafter, a measurable state 4 rate (V_4) (i.e., the rate after ADP phosphorylation) was monitored when a second pulse of ADP was added, but the phosphorylative cycle was soon inhibited before its completion by adding 1 μ g of oligomycin. Subsequently, a measurable oligomycin oxygen consumption rate (V_{olig}) was obtained after 1 μ M concentration of the uncoupling agent FCCP was added to obtain a rate of oxygen consumption in the absence of coupled oxidative phosphorylation (V_{FCCP}). Further, the oxygen consumption was recorded in presence of 15 mM succinate and 2.2 mM rotenone to elaborate the respiratory complex-II activity. Respiratory control ratios (RCR) and ADP/O were determined according to the standard protocol of Chance and Williams (1956).

Western blot analysis

For western blot analysis, rat PFC was collected and was lysed in buffer containing complete protease inhibitor cocktail. Protein concentrations were determined according to Bradford (1976). A standard plot was generated using bovine serum albumin. An aliquot of each sample were electrophoresed in 10% SDS-PAGE gels for GFAP, Iba-1, cytochrome-C, caspase-9 and caspase-3 proteins, transferred to polyvinylidene fluoride membranes and probed with specific antibodies. The membrane was incubated overnight with rabbit anti-GFAP (Abcam Plc., Cambridge, USA), anti-Iba-1 (Abcam Plc., Cambridge, USA), anti-cytochrome-C (Abcam Plc., Cambridge, USA), anti-caspase-9 (Abcam Plc., Cambridge, USA), and anti-caspase-3 (Abcam Plc., Cambridge, USA) polyclonal primary antibody at a dilution of 1:10000, 1:10000, 1:1000, 1:500, and 1:500 respectively. After detection with the desired antibodies against the proteins of interest, the membrane was stripped with stripping buffer (25 mM Glycine pH 2.0, 2% SDS for 30 min at room temperature) and re-probed overnight with rabbit anti β -actin (Santa Cruz Biotechnology Inc.; Santa Cruz, California, USA) polyclonal primary antibody at a dilution of 1:500 to confirm equal loading of protein. Further, membrane was probed with corresponding secondary antibodies. Immunoreactive band of proteins were detected by chemiluminescence using enhanced chemiluminescence

(ECL) reagents (Amersham Bioscience, USA). Quantification of the results was performed by densitometric scan of films. The immunoreactive area was determined by densitometric analysis using Biovis gel documentation software.

Data analysis

All the data were mean $\pm$ standard error of the mean (SEM). The escape latency of the animals on day-1 to 4 during MWM test paradigm was analyzed using repeated measure two-way analysis of variance (ANOVA) followed by Bonferroni Post-hoc test. All other statistical analyses of data were performed using one-way ANOVA with Newman-Keuls Post-hoc test to monitor significance among groups. $P < 0.05$ was considered as significant.

Results

OB (100 and 200 $\mu\text{L/kg}$) attenuated EB-induced memory deficits in MWM task in rats

Figure 2 illustrates the effect of OB (50, 100 and 200 $\mu\text{L/kg}$) on spatial learning ability in terms of the period of escape latency in Day-10 to Day-14 (A) and on spatial memory ability in terms of the time spent in the target quadrant (B), percentage of total distance travelled in target quadrant (C), and swimming speed of the animals in Day-14 of the MWM test protocol. However, OB (50 $\mu\text{L/kg}$) did not cause any significant change in the EB-

get quadrant (C) and swimming speed of the EB-challenged animals (D) in Day-14 of MWM test protocol. Statistical analysis revealed that there were significant differences in the escape latency among group [$F(6, 140) = 103.0$, $P < 0.05$] and day [$F(3, 140) = 356.3$, $P < 0.05$]. Further, there was a significant interaction between group and day in the escape latency of the animals [$F(18, 140) = 10.0$, $P < 0.05$]. Post-hoc test showed that there were no significant differences in escape latency of the animals among group on Day-10. OB (100 and 200 $\mu\text{L/kg}$) and UA significantly reduced the EB-challenged increase in the escape latency of the animals. The observations reported on Day-11 were consistent on Day-12 and 13 of the MWM test protocol.

Statistical analysis revealed that there were significant differences in the time spent in the target quadrant [$F(6, 35) = 56.5$, $P < 0.05$] and percentage of total distance travelled in target quadrant [$F(6, 35) = 28.8$, $P < 0.05$] of the animals in Day-14 of MWM test paradigm. However, there were no significant difference in swimming speed [$F(6, 35) = 0.2$, $P > 0.05$] of the animals among groups. Post-hoc test showed that OB (100 and 200 $\mu\text{L/kg}$) and UA significantly attenuated the EB-induced decrease in the amount of time spent and percentage of total distance travelled in target quadrant of the animals in Day-14 of the MWM test protocol. However, OB (50 $\mu\text{L/kg}$) did not cause any significant change in the EB-

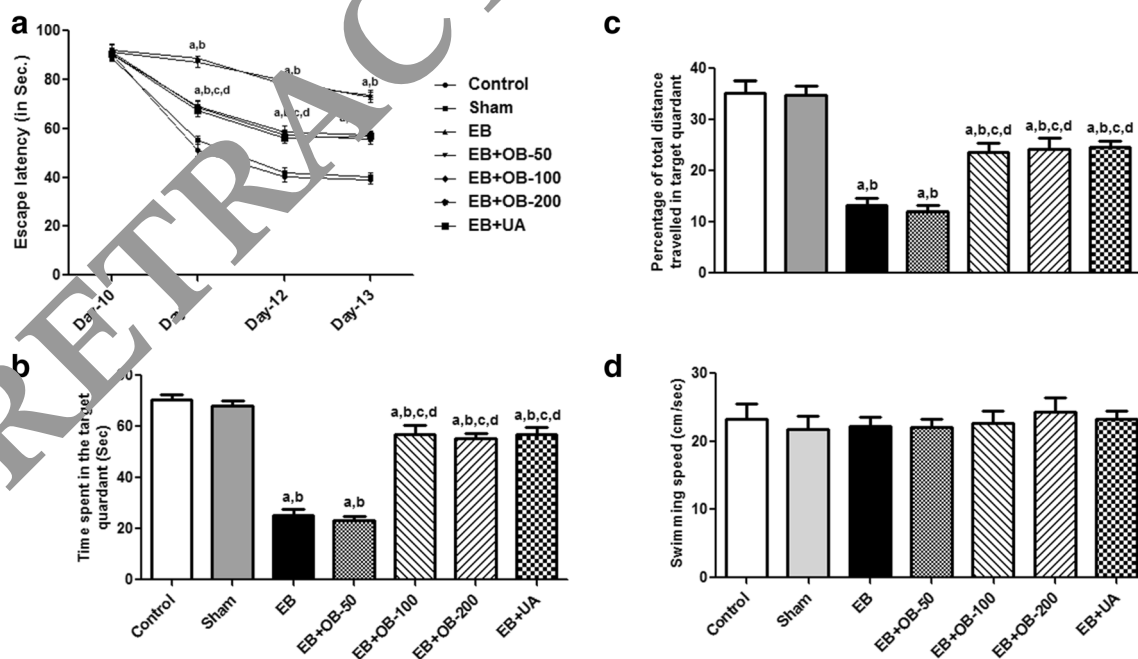


Fig. 2 Effect of OB (50, 100 and 200 $\mu\text{L/kg}$) on ethidium bromide (EB)-induced changes in the period of escape latency from Day-1 to Day-4 (a), the time spent in the target quadrant (b), percentage of total distance travelled in target quadrant (c) and swimming speed of the animals (d) in Day-5 of Morris water maze test protocol. All values are Mean $\pm$ Standard Error of Mean (SEM; $N = 6$). ^a $P < 0.05$ compared to Control,

^b $P < 0.05$ compared to Sham, ^c $P < 0.05$ compared to EB, ^d $P < 0.05$ compared to EB + OB-50, ^e $P < 0.05$ compared to EB + OB-100 and ^f $P < 0.05$ compared to EB + OB-200 (Two-way ANOVA followed by Bonferroni Post-hoc test for period of escape latency from Day-1 to Day-4 and for other parameters one-way ANOVA followed by Student Newman Keuls Post-hoc test)

induced decrease in the amount of time spent and percentage of total distance travelled in target quadrant of the animals in this test.

OB (100 and 200 $\mu\text{L/kg}$) attenuates EB-induced working memory deficits in Y-maze test in rats

Effect of OB (50, 100 and 200 $\mu\text{L/kg}$) on spatial memory in EB-challenged animals in Y-maze test protocol is illustrated in Fig. 3. Statistical analysis revealed that there were significant differences in the SAB [F (6, 35) = 16.2, $P < 0.05$] among groups. However, there were no significant differences in total arm entries [F (6, 35) = 0.4, $P > 0.05$] of the animals among groups. Post-hoc test showed that OB (100 and 200 $\mu\text{L/kg}$) and UA significantly increased the EB-challenged decrease in the SAB of the animals in the Y-maze test paradigm. However, OB (50 $\mu\text{L/kg}$) did not cause any significant change in the EB-induced decrease in the SAB of the animals in this test.

OB (100 and 200 $\mu\text{L/kg}$) attenuates EB-induced increase in the levels of IL-6 and TNF- α in rat PFC

Figure 4 depicts the effect of OB (50, 100 and 200 $\mu\text{L/kg}$) on the levels of IL-6 (A) and TNF- α (B) in EB challenged rat PFC. Statistical analysis revealed that there were significant differences in the levels of IL-6 [F (6, 35) = 11.5, $P < 0.05$] and TNF- α [F (6, 35) = 37.9, $P < 0.05$] among groups. Post-hoc analysis showed that OB (100 and 200 $\mu\text{L/kg}$) and UA significantly attenuated the EB-induced increase in the levels of IL-6 and TNF- α in rat PFC. OB (50 $\mu\text{L/kg}$) did not alter the EB-induced increase in the levels of IL-6 and TNF- α in rat PFC.

OB (100 and 200 $\mu\text{L/kg}$) attenuates EB-induced increase in the astrogliosis in rat PFC

Effect of OB (50, 100 and 200 $\mu\text{L/kg}$) on the extent of astrogliosis in terms of the levels of expression of GFAP (B) and Iba-1 (C) in PFC of EB challenged rats is

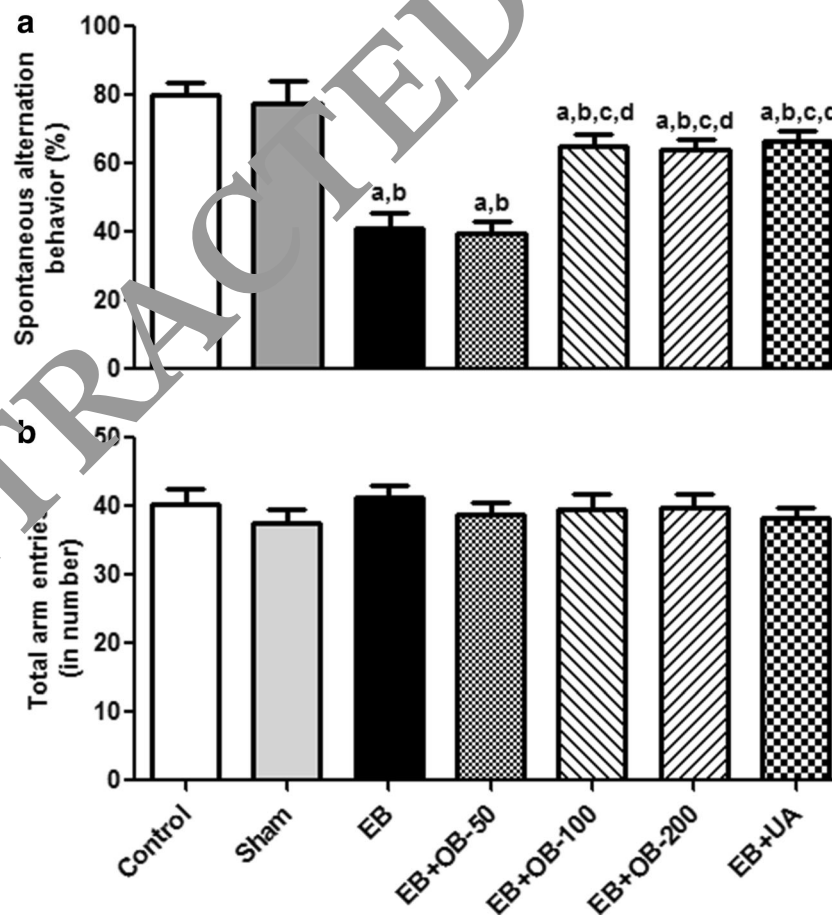


Fig. 3 Effect of OB (50, 100 and 200 $\mu\text{L/kg}$) on EB-induced changes in spatial memory in terms of spontaneous alternation behavior (SAB; **a**) and total arm entries (**b**) of the animals in Y-maze test paradigm. All values are Mean $\pm$ SEM ($N = 6$). ^a $P < 0.05$ compared to Control, ^b $P <$

0.05 compared to Sham, ^c $P < 0.05$ compared to EB, ^d $P < 0.05$ compared to EB + OB-50, ^e $P < 0.05$ compared to EB + OB-100 and ^f $P < 0.05$ compared to EB + OB-200 (One-way ANOVA followed by Student Newman Keuls Post-hoc test)

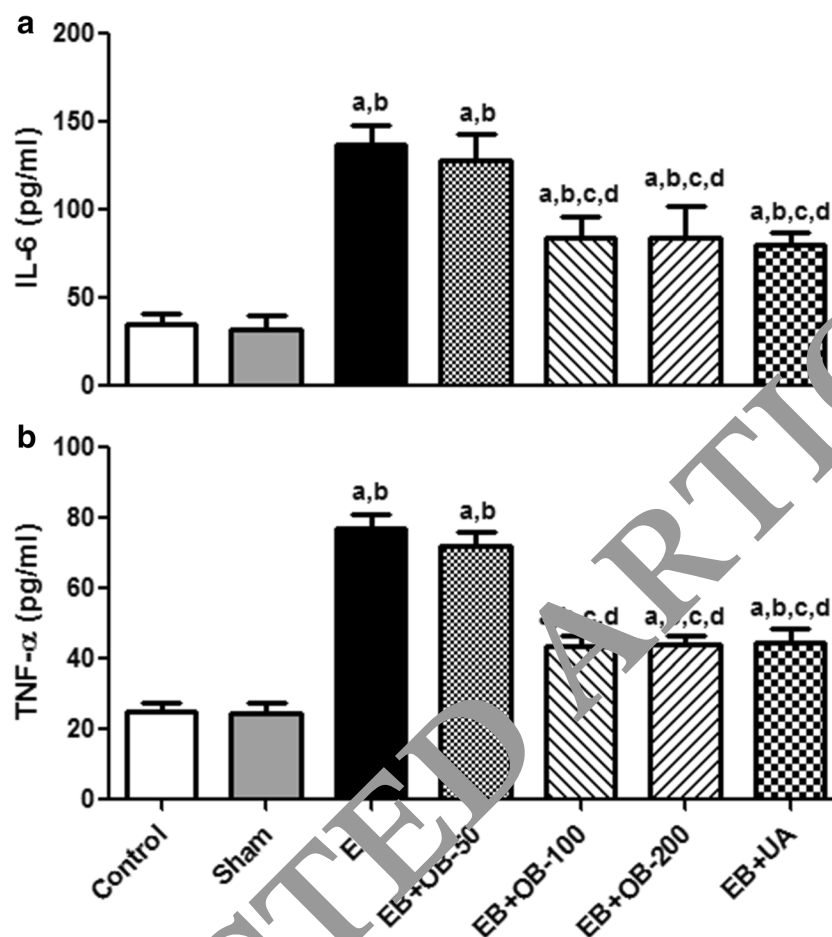


Fig. 4 Effect of OB (50, 100 and 200 μ L/kg) on EB-induced changes in the extent of neuroinflammation in terms of the levels of TNF- α (a) and IL-6 (b) in the rat pre-frontal cortex (PFC). All values are Mean $\pm$ SEM ($N=6$). ^a $P < 0.05$ compared to Control, ^b $P < 0.05$ compared to Sham, ^c $P < 0.05$ compared

to EB, ^d $P < 0.05$ compared to EB + OB-50, ^e $P < 0.05$ compared to EB + OB-100 and ^f $P < 0.05$ compared to EB + OB-200 (One-way ANOVA followed by Student Newman Keuls Post-hoc test)

depicted in Fig. 5. Statistical analysis revealed that there were significant differences in the levels of expression of GFAP [$F(6, 35) = 3.3, P < 0.05$] and Iba-1 [$F(6, 35) = 21.7, P < 0.05$] among groups. Post-hoc analysis showed that OB (100 and 200 μ L/kg) and UA significantly attenuated the EB-induced increase in the astrogliosis in rat PFC. OB (50 μ L/kg) did not alter the EB-induced increase in the astrogliosis in rat PFC.

OB (100 and 200 μ L/kg) attenuates EB-induced decrease in mitochondrial function and integrity in rat PFC

Figure 6 illustrates the effect of OB (50, 100 and 200 μ L/kg) on the level of formazan produced in MTT assay (A) and the fluorescence intensity of TMRM (B) in PFC of EB challenged rats. Statistical analysis revealed that there were significant differences in mitochondrial function [$F(6, 35) = 14.1, P < 0.05$] and integrity [$F(6, 35) = 11.3, P < 0.05$] among groups. Post-hoc analysis showed that OB (100 and 200 μ L/kg)

kg) and UA significantly attenuated the EB-induced decrease in the mitochondrial function and integrity in rat PFC. OB (50 μ L/kg) did not alter the EB-induced decrease in the mitochondrial function and integrity in rat PFC.

OB (100 and 200 μ L/kg) attenuates EB-induced decrease in mitochondrial RCR and ADP/O in rat PFC

Effect of OB (50, 100 and 200 μ L/kg) on the mitochondrial RCR (A) and ADP/O (B) in rat PFC of EB challenged rats is illustrated in Fig. 7. Statistical analysis revealed that there were significant differences in mitochondrial RCR [$F(6, 35) = 38.4, P < 0.05$] and ADP/O [$F(6, 35) = 18.2, P < 0.05$] among groups. Post-hoc analysis showed that OB (100 and 200 μ L/kg) and UA significantly attenuated the EB-induced decrease in the mitochondrial RCR and ADP/O in rat PFC. OB (50 μ L/kg) did not alter the EB-induced decrease in the mitochondrial RCR and ADP/O in rat PFC.

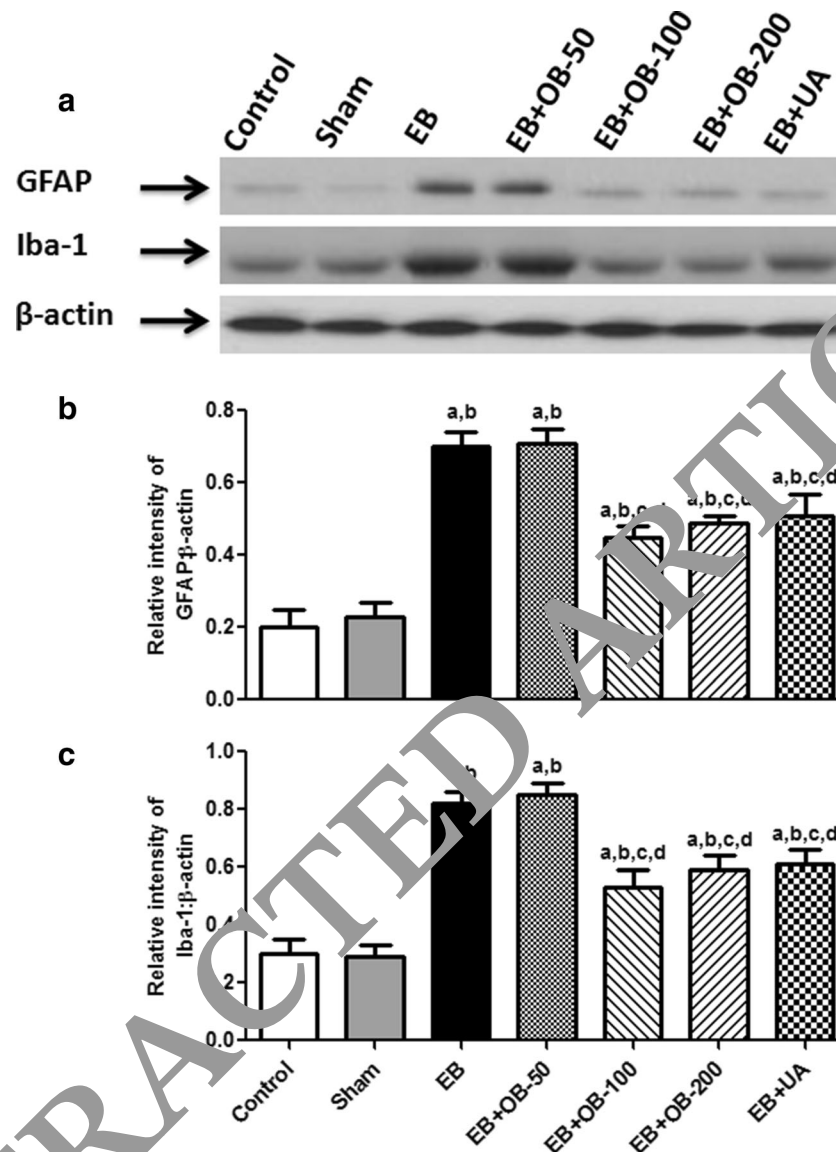


Fig. 5 Effect of OB (50, 100 and 200 µL/kg) on EB-induced changes in the extent of astrogliosis in terms of the levels of GFAP (a) and Iba-1 (b) in the rat PFC. All values are mean $\pm$ SEM ($N=6$). ^a $P < 0.05$ compared to Control, ^b $P < 0.05$ compared to Sham, ^c $P < 0.05$ compared to EB, ^d $P <$

0.05 compared to EB + OB-50, ^e $P < 0.05$ compared to EB + OB-100 and ^f $P < 0.05$ compared to EB + OB-200 (One-way ANOVA followed by Student Newman Keuls Post-hoc test)

OB (100 and 200 µL/kg) attenuates EB-induced increase in mitochondria-dependent apoptosis in rat PFC

Figure 8 depicts the effect of OB (50, 100 and 200 µL/kg) on EB-induced mitochondria-dependent apoptosis in terms of the levels of expression of cytochrome-C (B), caspase-9 (C) and caspase-3 (D) in rat PFC. Statistical analysis revealed that there were significant differences in the levels of expression of cytochrome-C [$F(6, 35) = 59.7$, $P < 0.05$], caspase-9 [$F(6, 35) = 33.1$, $P < 0.05$] and caspase-3 [$F(6, 35) = 37.5$, $P < 0.05$] among groups. Post-hoc analysis showed that OB (100 and 200 µL/kg) and

UA significantly attenuated the EB-induced increase in the mitochondria-dependent apoptosis in rat PFC. OB (50 µL/kg) did not alter the EB-induced increase in the mitochondria-dependent apoptosis in rat PFC.

Discussion

The present study for the first time demonstrates that OB (100 and 200 µL/kg) and UA attenuated EB-induced cognitive impairments in the animals. Further, OB (100 and 200 µL/kg) and UA attenuated EB-induced neuroinflammation, astrogliosis and mitochondrial dysfunction in PFC

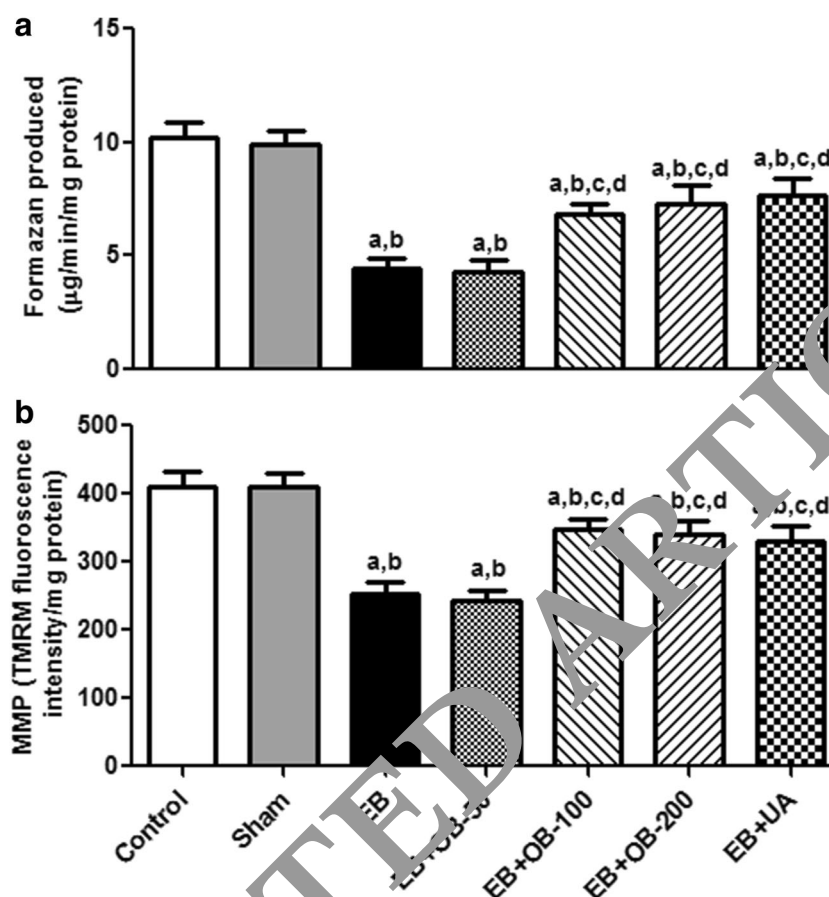


Fig. 6 Effect of OB (50, 100 and 200 μL/kg) on EB-induced changes in the mitochondrial function in terms of the level of formazan produced in MTT assay (**a**) and integrity in terms of the fluorescence intensity of TMRM (**b**) in rat PFC. All values are Mean ± SEM (n = 6). ^aP < 0.05

compared to Control, ^bP < 0.05 compared to Sham, ^cP < 0.05 compared to EB, ^dP < 0.05 compared to EB + OB-50, ^eP < 0.05 compared to EB + OB-100 and ^fP < 0.05 compared to EB + OB-200 (One-way ANOVA followed by Student Newmann Keuls Post-hoc test)

of the animals. Thus, it can be assumed that OB could be an alternative option in the pharmacotherapy of MS.

The loss in the explicit memory is well established in the pathophysiology of MS (Feinstein 2015; D'Amico et al. 2016; González Torrealba et al. 2017). In the present study, EB caused a significant reduction in spatial and working memory in the MWM and Y-maze test paradigms similar to that of earlier observations (Ghaffari et al. 2015; Goudarzvand et al. 2015). Moreover, the analyses of swimming velocity to reach the hidden platform in MWM and total number of entries in Y-maze revealed no differences between experimental groups in the present study. These results demonstrate that the deleterious effects of EB microinjection on spatial learning and memory are not due to any non-specific fluctuations in gross motor activity or motivational state. Pre-treatment of OB (100 and 200 μL/kg) attenuated EB-induced loss in memory formation in the animals. In consistent to our results, it has been reported that OB attenuated memory loss in the animals of the different experimentally-induced neurological disorders (Singh et al. 2018; Ayuob et al. 2018). Moreover, UA improved the memory formation in the

animals subjected to EB injection in the present study similar to that of earlier study (Liang et al. 2016).

The present study documented that EB caused neuroinflammation in terms of the elevated levels of pro-inflammatory cytokinins such as IL-6 and TNF-α in the PFC of the animals. Similar to our findings, the phenomenon of neuroinflammation is well reported in the pathogenesis of EB-induced dementia in the animals (Carvalho et al. 2015). Interestingly, this is the first report that EB caused a significant increase in the process of astrogliosis in terms of increased levels of GFAP and Iba-1 in the PFC of the rodents. Moreover, OB (100 and 200 μL/kg) and UA significantly attenuated EB-induced increase in the astrogliosis in the rat brain region. These observations indicate that OB (100 and 200 μL/kg) may exhibit neuroprotective activity against EB-induced loss in the memory formation possibly through modulatory effect on neuroinflammation and astrogliosis in the PFC of the animals.

It is reported that mitochondrial dysfunction is common to astrogliosis and neuroinflammation (Niinobu et al. 2000; Lovatt et al. 2007). In the present study,

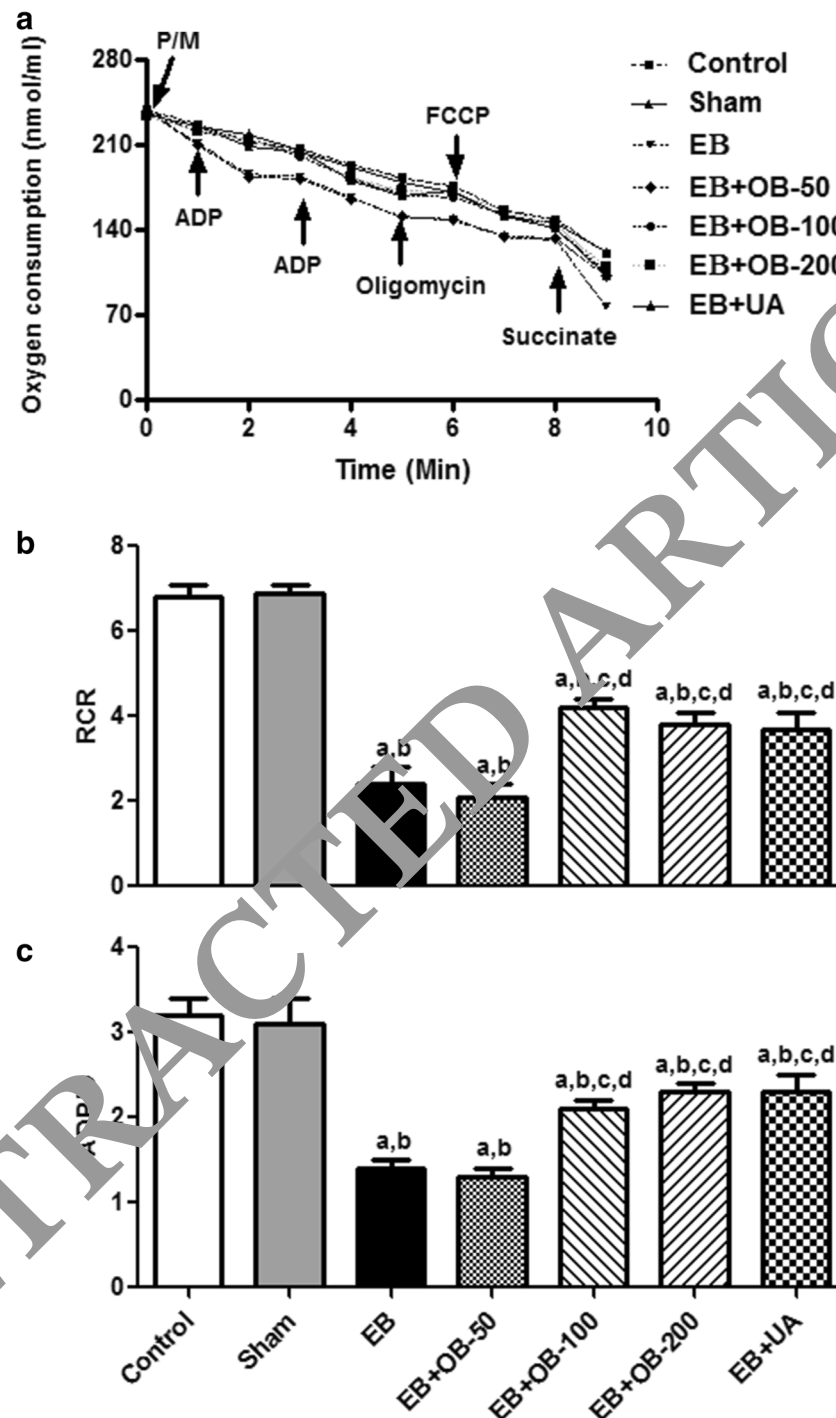


Fig. 7 Representative Oxygraph of PFC (**a**) of an animal in each group of the experimental protocol to show the level of oxygen consumption in different stages of mitochondrial respiration. Effect of OB (50, 100 and 200 $\mu\text{L/kg}$) on EB-induced changes in the mitochondrial respiratory control ratio (RCR; **b**) and ADP/O (**c**) in rat PFC is depicted. All values are

Mean $\pm$ SEM ($N = 6$). ^a $P < 0.05$ compared to Control, ^b $P < 0.05$ compared to Sham, ^c $P < 0.05$ compared to EB, ^d $P < 0.05$ compared to EB + OB-50, ^e $P < 0.05$ compared to EB + OB-100 and ^f $P < 0.05$ compared to EB + OB-200 (One-way ANOVA followed by Student Newman Keuls Post-hoc test)

mitochondrial dysfunction in terms of decreased mitochondrial function and integrity in the PFC of the EB challenged animals. Further, mitochondrial bioenergetics in terms of mitochondrial oxygen consumption was

observed in the PFC of the rodents challenged to EB. Additionally, EB increased the mitochondria-dependent apoptotic markers such as the levels of expression of cytochrome-c, caspase-9 and caspase-3 proteins in the rat

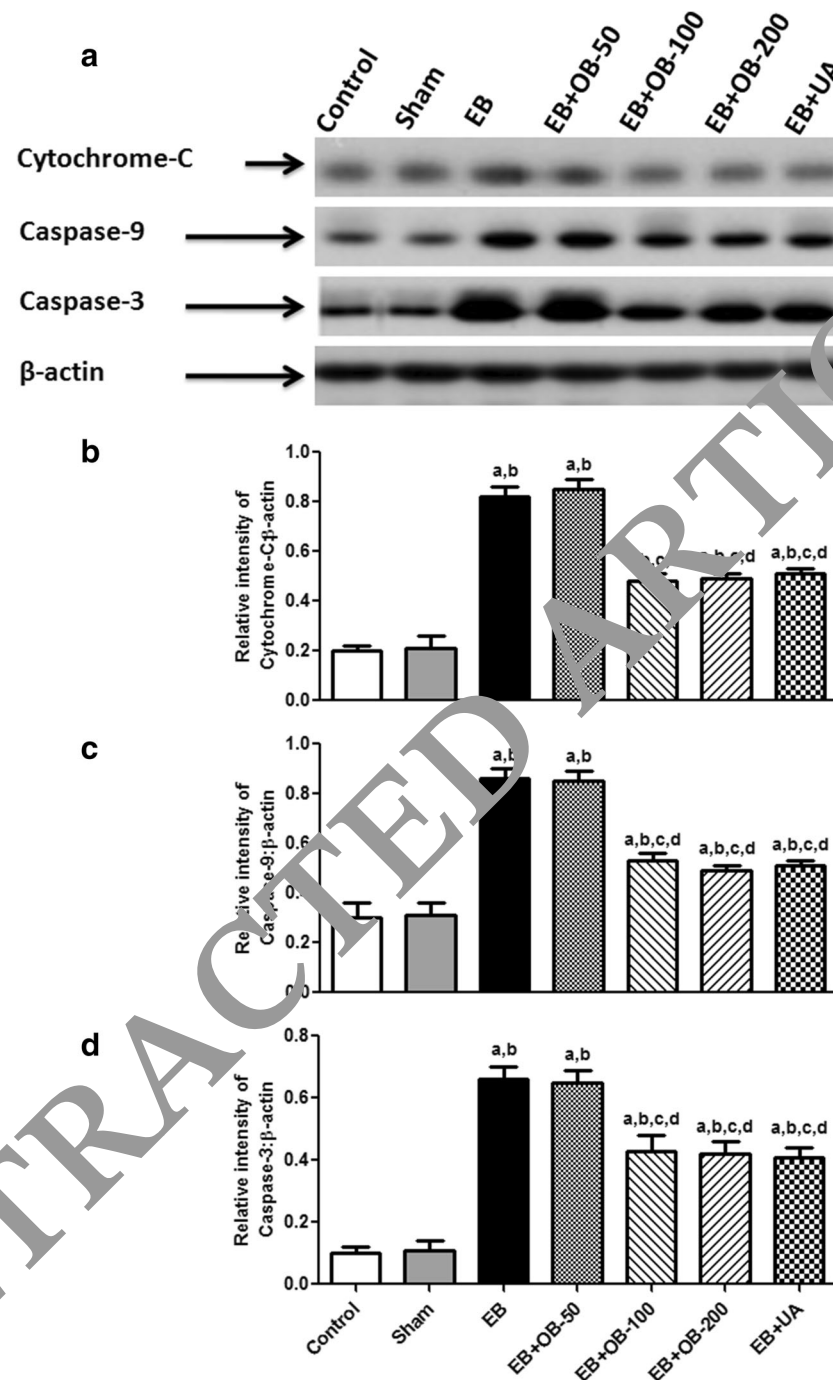


Fig. 8 Effect of OB (50, 100 and 200 $\mu\text{L/kg}$) on EB-induced changes in the levels of expression of Cytochrome-C (b), Caspase-9 (c) and Caspase-3 (d) in rat PFC. The blots are representative of Cytochrome-C, Caspase-9 and Caspase-3 (a) in PFC. The results in the histogram are expressed as the ratio of relative intensity of levels of expression of Cytochrome-C,

Caspase-9 and Caspase-3 to β -actin. All values are Mean $\pm$ SEM ($N = 6$). ^a $P < 0.05$ compared to Control, ^b $P < 0.05$ compared to Sham, ^c $P < 0.05$ compared to EB, ^d $P < 0.05$ compared to EB + OB-50, ^e $P < 0.05$ compared to EB + OB-100 and ^f $P < 0.05$ compared to EB + OB-200 (One-way ANOVA followed by Student Newman Keuls Post-hoc test)

PFC. All these observations indicate that EB induced mitochondrial dysfunction in the PFC of the animals. Moreover, OB (100 and 200 $\mu\text{L/kg}$) and UA significantly attenuated EB-induced mitochondrial dysfunction in the rat PFC. It is interesting to note that UA improves

mitochondrial function in brain region of the animals in neurological disorders (Shih et al. 2004). Hence, UA in OB could be responsible in the attenuation of the mitochondrial dysfunction, neuroinflammation and astrogliosis induced by EB in the animals.

In conclusion, OB (100 and 200 $\mu\text{L/kg}$) attenuated EB-induced cognitive impairment in the animals. Further, OB (100 and 200 $\mu\text{L/kg}$) attenuated EB-induced neuroinflammation, astrogliosis and mitochondrial dysfunction in the PFC of the animals. Thus, it can be assumed that OB could be a potential alternative drug candidate in the pharmacotherapy of MS.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interests.

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